

Information Letter

Building Together: A Gamification Framework for Real-Time Casual Social Interaction Among Generation Z in the Mixed Reality Metaverse

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You are invited to participate in this study.

Background

The rapid development of immersive technologies—particularly Mixed Reality (MR), which seamlessly blends physical and virtual environments—has created new opportunities for social interaction within the Metaverse. MR enables real-time, spontaneous, and embodied communication, making it a promising space for meaningful digital engagement.

Meanwhile, gamification—the application of game-like elements in non-game contexts—has proven to be a powerful method for increasing motivation and engagement in education, workplaces, and digital platforms. Prior studies have shown that game mechanics such as avatars, identity expression, achievement systems, and social collaboration can foster immersive and rewarding interactions in virtual spaces.

However, current Metaverse platforms often focus on Virtual Reality (VR) and game-centric models, offering limited support for organic, casual social interaction. There is also limited research on how these interactions unfold in MR environments, especially among Generation Z—digital natives who value self-expression, instant feedback, and immersive connectivity.

Our study aims to address these gaps by exploring how gamification can enhance Generation Z's social interaction in Mixed Reality environments. To do this, we are developing and testing a new design model—the Affordance-Driven Gamification Framework (ADGF)—to support real-time, casual, and inclusive Metaverse experiences.

We invite you to take part in an experimental prototype experience over the next few weeks (dates to be confirmed) to help us evaluate and refine this gamification framework. The project is supervised by Associate Professor Mohd Fairuz Shiratuddin, Dr Mostafa Hamadi, Dr Joo Yeon Park, and Dr Thao Duong, and has been reviewed to ensure it meets all ethical standards.

Aim of the Study

This study aims to explore how gamification can enhance casual, real-time social interaction among Generation Z users within Mixed Reality (MR) Metaverse environments. Specifically, it investigates which game mechanics—such as avatar identity, gesture-based collaboration, and immersive feedback—can foster deeper engagement and spontaneous connection. To support this, the study introduces and tests the Affordance-Driven Gamification Framework (ADGF), a design model developed to align game elements with the social and technological affordances of MR platforms. Through participant feedback and interaction with apps, this research seeks to refine the framework and offer practical design guidelines for building socially engaging, inclusive MR Metaverse experiences tailored to the behavioral preferences of Gen Z.

What Does Your Participation Involve?



If you choose to participate in this study, you will be invited to take part in a short interactive session using a Meta Quest headset. During this session, you will explore a selected Mixed Reality (MR) application or game designed to encourage spontaneous, casual social interactions in an immersive environment. As a participant, you will be asked to:

- Complete a brief pilot survey collecting demographic information (e.g., age, gender), your preferred games and social apps, and any conditions that may be negatively affected by immersive visual technologies. This step will be completed online and should take about **5 minutes**. It will not take much of your time unless you wish to share additional information about your preferences. In addition, we will send you the links to the apps that will be used in the experiment. If you have sufficient time, you may review them to get some basic background information; however, this is optional.
- Attend an introductory session in a designated room on campus. This session will take approximately **5-15** minutes and will introduce you to the device and the apps used in the study.
- Use the MR app or game for about **25-30 minutes** in a supervised, campus-based setting alongside other participants. During the experiment, you will interact with virtual elements and, in some cases, engage with other participants in a multi-player MR environment. There are no strict rules; you are simply encouraged to interact with the MR environment as much as possible-earning points, exploring, and enjoying the experience.
- Participate in a guided group discussion immediately following the session (approximately **20-25** minutes). You will be encouraged to share your experiences of the MR interaction in a safe and respectful environment together with other participants. The investigator will be present to facilitate and guide the discussion.

With your consent, all sessions will be audio recorded and video recorded for research purposes (including screen capture during the MR experience). All data collected will be securely stored and used solely for the purposes of this study. Your responses will remain anonymous, and no personally identifying information will appear in any publications or reports.

In accordance with Murdoch University's ethics guidelines, all data will be stored on a password-protected university server and securely deleted five years after the completion of the study.

Voluntary Participation and Withdrawal from the Study

It is important that you understand your involvement in this study is entirely voluntary. While we would be pleased to have you participate, we fully respect your right to decline. Choosing not to take part will have no consequences for you and will not affect any services or support you may receive from the university or elsewhere. If you choose to participate, you may withdraw from the study at any time without needing to provide a reason. Should you decide to withdraw, all data you have provided will be securely destroyed. Please note, however, that if your data has already been de-identified (i.e., all personal identifiers have been removed) or included in published findings, it may not be possible to remove it from the study at that stage. In such cases, any data used will remain completely anonymous and untraceable to you.

Your privacy

Your privacy is very important to us. Your participation in this study and any information will be treated in a confidential manner. Your name and identifying details will not be used in any publication arising out of the research.

Possible Benefits

While there may be no direct personal benefit to you from participating in this study, your involvement will contribute to important research on how gamification and Mixed Reality (MR) technologies can support meaningful, casual social interactions among Generation Z users in immersive environments.

By participating, you may:

- Gain firsthand experience with emerging MR technologies and social interaction designs.
- Reflect on your own social preferences and engagement styles in digital environments.
- Contribute valuable feedback that will help shape future MR experiences and gamified platforms in the Metaverse.



- Help researchers and designers develop more inclusive, engaging, and user-centered immersive environments for future generations.
- You will also have the opportunity to explore the XR world as a pioneer in this emerging field, experiencing firsthand some of the most advanced XR technologies currently available—including the Meta Quest 3 and Apple Vision Pro - both during and after the experimental session briefly.

Your participation may also support broader advancements in immersive technology, education, and digital social design.

Possible Risks

There are no specific risks anticipated with participation in this study. However, if you find that you are becoming distressed or overwhelmed during the Mixed Reality experiences such as due to motion discomfort, sensory overload, or social anxiety, you will be advised to receive support from the university's wellbeing services. Alternatively, we will arrange for you to see a counsellor at no expense to you.

Reimbursement

To thank you for your time and contribution, you will be entered into a draw to win a \$100 gift card after completing the session. A total of three \$100 gift cards will be awarded. This is offered as a token of appreciation for your participation in the study, not as an inducement.

Questions

If you would like to discuss any aspect of this study, please feel free to contact either Kris Xie via email at Kris.Xie@murdoch.edu.au or Associate Professor Mohd Fairuz Shiratuddin via email at m.fairuz@murdoch.edu.au. Either of us would be happy to discuss any part of the research with you.

Once we have analysed the information from this study, we will email a summary of our findings. You can expect to receive this feedback in 3 weeks.

We would like to thank you in advance for your assistance with this research project. We look forward to hearing from you soon.

If you are happy to participate in the study, please contact Kris via email at Kris.Xie@murdoch.edu.au (Kris Xie) to arrange a suitable time for your participation session.

This study has been approved by the Murdoch University Human Research Ethics Committee (Approval xxxx/xxx). If you have any reservation or complaint about the ethical conduct of this research, and wish to talk with an independent person, you may contact Murdoch University's Research Ethics & Integrity on Tel. 08 9360 6677 (+61 8 9360 6677 for overseas studies) or e-mail human.ethics@murdoch.edu.au. Any issues you raise will be treated in confidence and investigated fully, and you will be informed of the outcome.